

Microsoft Cognitive Services For Accessibility

2022 // Dennie Declercq



Let's introduce me



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Content

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- Speech
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What is Microsoft Cognitive Services?



Global picture

- Microsoft AI solution for coders
- No need to be a data scientist
- Accessible via easy to use API's in modern languages.
- Easy to integrate into day to day applications
- There are tons of Cognitive Services
- Still growing
- Possibility to mix and match different services



Speech



Global picture

- Speech Synthesis
- Speech interpretation
- Language features
- Custom speech



Use Cases

- Read aloud documents
- Read aloud websites
- Click on an element to read aloud
- Speak with the voice of a parent or caregiver
- Confirm written text
- Is used separately on clickable elements
- Is used in screenreaders
- Control device via voice commands



Example

- Microsoft Translator:
 - <https://www.bing.com/translator>
 - 'Translator' app



Vision



Global picture

- Picture analysis
- Picture recognition
- Custom pictures
- Video analysis
- Anomaly detection



Use Cases

- Analyze user's emotions
- Analyze physical money
- Analyze clothes of a person to chose clothes for the person
 - Based on weather
 - Based on agenda
- Detect people in spaces and tell them if they are alone or not alone
- Analyze menu card in restaurant (and combine with text-to-speech)
- Help with counting objects (analyze objects to count, ask count to end-user, compare count).
- Assist with shaving or hair dress



Example

- Seeing AI: <https://news.microsoft.com/videos/seeing-ai-app-microsoft>



Immersive Reader



Global picture

- Tools to make text documents easier to understand by end user
- Tools to reshape text (colorize, add space, ...)
- Link pictographs to text
- Read aloud text



Use Cases

- Help with recipes on cooking websites
- Help with school documents for students with dyslexia
- Assist with unknown word-usage on websites



Example

- Office 365 & Immersive reader: <https://content.cloudguides.com/en-us/guides/Help%20students%20read%20more%20effectively>



Resources

- <https://www.microsoft.com/en-us/ai/ai-for-accessibility>
- <https://docs.microsoft.com/en-us/learn/paths/accessibility-fundamentals/>
- <https://accessibledreams.net/>
- <https://www.ddsoft.be/en/we-connect/?lang=en>
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Q&A

